LCM and HCF

LCM (2, 3) = 6 → smallest no. which is a multiple of both.

HCF (12, 15) = 3 → biggest no. which can divide both numbers.

CONCEPT

L.C.M and HCF Important Formulas

- Product of two numbers (First number x Second Number) = H.C.F. X L.C.M.
- H.C.F. of a given number always divides its L.C.M.
- To find the **greatest number** that will exactly divide x, y and z. Required number = **HCF** of x, y and z
- To find the **Largest number** that will divide x, y and z leaving remainders a, b and c respectively. Required number = **HCF** of (x -a), (y- b) and (z – c)
- To find the least number which is exactly divisible by x, y and z. Required number = LCM of x, y and z
- To find the **least number** which when divided by x, y and z leaves the remainders a, b and c respectively. It is always observed that, (x – a) = (y – b) = (z- c) = K (say). Required number = (**LCM** of x, y and z) – K.
- To find the least number which, when divided by x, y and z leaves the same remainder r in each case. Required number = (LCM of x, y and z) + r
- To find the greatest number that will divide x, y and z leaving the same remainder 'r' in each case. Required number = HCF of (x -r), (y- r) and (z- r)
- Largest number which divides x, y, z to leave same remainder = H.C.F. of (y-x), (z-y), (z-x).
- HCF of two prime numbers is always 1.

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1. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

A. 4 ✓ B. 7 C. 9 D. 13

91−43, 183−91, 183−43

4 | 48, 92, 140

12, 23, 35

2. The H.C.F. of two numbers is 23 and the other two factors of their L.C.M. are 13 and 14. The larger of the two numbers is:

A. 276 B. 299 C. 322 ✓ D. 345

A = HCF × x = 23×13

B = HCF × y = 23×14 = 322

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3. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?

A. 4 B. 10 C. 15 D. 16

→ har bell 2,4,6 etc second me baj rahi apan ko nikaalna hai ki 30 min me ye bells ne kitni baar saath me ring kiya. (which will be the LCM).

LCM

2 | 2, 4, 6, 8, 10, 12

2 | 1, 2, 3, 4, 5, 6

3 | 1, 1, 3, 2, 5, 3

1, 1, 1, 2, 5, 1

LCM = 2×2×3×2×5 = 120 sec = 2 min

$\frac{30}{2} = 15$

1+15 = 16

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4. Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is:

A. 4 ✓ B. 5 C. 6 D. 8

4665−1305, 6905−4665, 6905−1305

10 | 3360, 2240, 5600

4 | 336, 224, 560

4 | 84, 56, 140

7 | 21, 14, 35

3, 2, 5

HCF = 10×4×4×7 = 1120

1+1+2+0 = 4

5. Three numbers are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:

A. 40 ✓ B. 80 C. 120 D. 200

3x 4x 5x

HCF = x

LCM = HCF × Rem Factors

2400 = x × 3×4×5

$x = \frac{\cancel{2400}^{120}}{3\times\cancel{4\times5}} = 40$

6. Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11 respectively.

A. 20   ✓ B. 12   C. 8   D. 48

148 − 4, 246 − 6, 623 − 11

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\begin{array}{c|l}
4 & 144, 240, 612 \\ \hline
3 & 36, 60, 153 \\ \hline
 & 12, 20, 51
\end{array}
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$$\underline{\underline{4 \times 3 = 12}}$$